



Mediacrest Training College

Nairobi, Kenya

Advanced Data Analytics & Machine Learning

8-Week Advanced Training Program

Capstone Project Documentation

E-Commerce Analytics & Customer Churn

OmniChannel Retail Strategy & Predictive Modeling

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Role: Statistician, Data Analyst & ML Expert

Session Date: 31st August 2026

Mode: Independent Capstone Execution

“A model is only as valuable as the business decisions it informs. Analytics must bridge the gap between raw data and strategic action.”

Project Scenario: OmniChannel Retail & Subscription Analytics

You have been hired as the **Lead Data Analyst** for a rapidly growing Kenyan digital lifestyle brand. The company operates a dual-revenue model: an e-commerce platform selling physical goods and a premium subscription service for digital content.

The executive team is struggling with two critical issues:

1. **Marketing Inefficiency:** Despite high web traffic, conversion rates and revenue per session vary wildly across channels (TikTok, Email, Google, etc.) and devices. The marketing budget is not optimized for ROI.
2. **Subscription Churn:** The company is losing subscribers at an alarming rate. The product team lacks data-driven insights into which demographic or account factors drive users to cancel their subscriptions.

You are provided with two datasets:

- **Web-Analytics.csv:** Tracks user sessions, marketing channels, device types, landing pages, funnel stages (Awareness to Retention), and financial metrics.
- **churn_detection_data.csv:** Contains customer demographics, account balances, employment status, and a binary target variable indicating subscription churn.

Your mission is to clean, analyze, model, and translate these datasets into a strategic roadmap for the executive team.

Project Requirements

This capstone requires you to execute a comprehensive analytical pipeline covering data preparation, exploratory data analysis, rigorous statistical testing, predictive modeling, and business strategy formulation. You must complete all **10 Tasks** outlined in this document.

Academic Integrity Code Hints

This document intentionally contains **no Python code hints**. You are expected to independently research, write, and debug the Python code required to execute these tasks using libraries such as `pandas`, `numpy`, `scipy`, `statsmodels`, `scikit-learn`, `matplotlib`, and `seaborn`.

Contents

Project Scenario	1
1 Analytical Pipeline & Tasks	3
1.1 Data Importation and Preparation	3
1.2 Data Wrangling and Manipulation	3
1.3 Data Visualization	3
1.4 Hypothesis Testing	4
1.5 Correlation and Regression Analysis	4
1.6 Binary Logistic Regression & Model Evaluation	6
1.7 Business Strategy & Policy Recommendations	6
2 Deliverables & Submission Guidelines	8
2.1 Required Outputs	8
2.2 Evaluation Rubric	8

1 Analytical Pipeline & Tasks

1.1 Data Importation and Preparation

Before any analysis can begin, the raw data must be ingested and audited for quality.

Task 1: Data Importation and Initial Audit

1. Load both `Web-Analytics.csv` and `churn_detection_data.csv` into your analytical environment.
2. Perform an initial audit: display the dimensions, data types, and summary statistics for both datasets.
3. Identify missing values, duplicates, and structural anomalies.
4. Apply appropriate cleaning strategies (e.g., dropping irrelevant columns, handling missing numerical values with median/mean, and handling missing categorical values).

1.2 Data Wrangling and Manipulation

Raw data rarely contains the exact features needed for advanced modeling. Feature engineering is required.

Task 2: Data Wrangling and Feature Engineering

1. Standardize column names and convert date columns in the web analytics dataset to proper datetime objects.
2. **Web Analytics Engineering:** Create derived metrics such as Conversion Rate, Revenue per Session, and Return on Ad Spend (ROAS) where mathematically applicable.
3. **Churn Data Engineering:** Encode categorical variables (e.g., Gender, Education, Employment) appropriately for statistical modeling. Scale numerical features if necessary.

1.3 Data Visualization

Visualizations must tell a story to non-technical stakeholders.

Task 3: Exploratory Data Analysis (EDA) & Visualization

1. Visualize the distribution of traffic sources, devices, and landing pages.
2. Create a visualization showing user progression or conversion volumes across the funnel stages (Awareness, Consideration, Decision, Retention).
3. Plot the distribution of customer churn across key demographics (Age bins, Gender, Education, Marital Status).
4. Ensure all plots have professional titles, labels, and legends.

1.4 Hypothesis Testing

Statistical significance must be proven before making business claims.

Task 4: Parametric Hypothesis Testing

1. Formulate null (H_0) and alternative (H_1) hypotheses for the following scenarios:

• Is there a significant difference in the **mean revenue** generated between Desktop and Mobile users? (Use an Independent T-test).

• Does the **average session value** differ significantly across the top three traffic sources? (Use One-way ANOVA).

2. Check the assumptions for these tests (normality, homogeneity of variance).

3. Report the test statistics, p-values, and your business conclusions.

Task 5: Non-Parametric Hypothesis Testing

1. Formulate hypotheses for scenarios where parametric assumptions fail or data is ordinal:

• Does the distribution of **conversion counts** differ significantly across the funnel stages (Awareness, Consideration, Decision)? (Use the Kruskal-Wallis Test).

• Is there a significant difference in **customer account balance** between churned and retained customers? (Use the Mann-Whitney U Test).

2. Interpret the results and explain why a non-parametric test was appropriate for these specific variables.

1.5 Correlation and Regression Analysis

Understanding relationships between variables is key to predictive modeling.

Task 6: Correlation Analysis

1. Compute the Pearson and Spearman correlation matrices for numerical features in both datasets.

2. Identify highly correlated variables (multicollinearity) that might distort regression models.

3. Visualize the correlation matrix using a heatmap. Interpret the relationship between marketing spend, session metrics, and revenue.

Task 7: Multiple Linear Regression Analysis

1. Build a multiple linear regression model to predict a continuous financial metric (e.g., **Revenue**) using web analytics features.

2. Evaluate model assumptions: linearity, independence, homoscedasticity, and normality of residuals.

3. Interpret the coefficients, R-squared, and Adjusted R-squared. Discuss the business impact of the significant predictors.

1.6 Binary Logistic Regression & Model Evaluation

Predicting customer churn requires a robust classification model.

Task 8: Binary Logistic Regression Analysis

1. Using the churn dataset, build a binary logistic regression model to predict customer churn.

2. Select predictors based on your EDA and correlation findings.

3. Interpret the **Odds Ratios (OR)** for key variables. Explain what an $OR > 1$ and $OR < 1$ means in the context of customer retention.

The Prevalence Trap

If the churn rate in your dataset is high, remember that Odds Ratios (OR) will exaggerate the Risk Ratio (RR). Be prepared to explain this distinction to stakeholders.

Task 9: Model Evaluation & Diagnostics

1. Evaluate the logistic regression model using a confusion matrix.

2. Calculate and interpret Accuracy, Precision, Recall (Sensitivity), and F1-Score.

3. Plot the ROC curve, calculate the Area Under the Curve (AUC), and determine the optimal classification threshold for minimizing false negatives (missed churners).

Kenyan Market Context

In the Kenyan digital market, acquiring a new customer via mobile money or social media ads can cost 3x to 5x more than retaining an existing one. Therefore, optimizing for **Recall** (catching as many potential churners as possible) is often more critical than Precision in subscription models.

1.7 Business Strategy & Policy Recommendations

Data must translate into action.

Task 10: Business Strategy & Policy Memo

Synthesize your statistical findings into a 2-page strategic memo addressed to the Executive Team. Your memo must include:

1. **Marketing Optimization:** Recommend specific budget reallocations based on your ROAS, ANOVA, and regression findings. Which channels and devices should receive more investment?

2. **Churn Mitigation Policy:** Propose a targeted retention strategy based on the logistic regression risk factors. How should the product team intervene for high-risk demographics?

3. **Model Limitations:** Briefly acknowledge any data limitations or assumptions made

during your analysis.

2 Deliverables & Submission Guidelines

2.1 Required Outputs

1. **Jupyter Notebook:** A clean, well-commented, and logically structured notebook containing all Python code, outputs, and markdown explanations for Tasks 1 through 9.
2. **Executive Memo (PDF):** The strategic business memo requested in Task 10.
3. **Presentation Deck:** A 5-slide summary deck highlighting the most critical EDA findings, model performance, and strategic recommendations.

2.2 Evaluation Rubric

- **Technical Rigor (40%):** Correct application of statistical tests, regression diagnostics, and model evaluation metrics.
- **Code Quality (20%):** Clean, modular, and well-documented Python code.
- **Business Acumen (30%):** Depth of insights, quality of visualizations, and relevance of the strategic recommendations to the Kenyan market.
- **Communication (10%):** Clarity of the executive memo and presentation.

Pro Tip for Success

Do not just report numbers; tell a story. When presenting the logistic regression results, don't just say "Education has an OR of 1.4." Say, "Customers with secondary education are 40% more likely to churn, suggesting we need to introduce tiered pricing or localized content to retain this segment."

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